Musanze HarvestLink Cooperative - Staff Decision Dashboard

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Harvest and Dispatch Decision Dashboard - UI/UX design proposal

1. Operational problem

HarvestLink coordinates potato collection from several smallholder farms around Musanze. Before dispatch, the operations team must decide, for every consignment, how much produce to expect, whether it needs special handling, and which collection points are operationally similar. Today these three decisions are made from memory and paper records, so mistakes are only discovered after the truck has left.

2. Stakeholder

Primary stakeholder: the Dispatch Officer at the collection centre. They are not technical, they work under time pressure at the loading bay, and they need to justify each decision to the cooperative committee. Secondary stakeholders: the Collection Point Supervisor and the Cooperative Manager who reviews weekly performance.

3. The three decisions the dashboard supports

DECISION 1

How much will this consignment weigh?

A regression estimate of expected yield in kg, shown with an honest uncertainty band.

DECISION 2

Does this consignment need dispatch attention?

A yes/no flag with a probability, so the officer can see how close the case is to the line.

DECISION 3

Which collection points behave alike?

A cluster label that groups similar operating profiles for route and resource planning.

4. Group and members

Group code: AI-G15. Members and assigned roles: see AI_A1_G15_CONTRIBUTIONS.pdf. Dataset: lecturer-issued AI_A1_G15.csv, used unchanged.

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User journey and information flow

The same six stages repeat for every dispatch round. The human decision is always the last stage: the dashboard advises, the officer decides. Nothing is dispatched automatically.

1. **Load the round**
   Dispatch officer drops the day's CSV into the upload area, or the system picks up the scheduled file.
2. **Check the data**
   The dashboard reports row count, missing cells and the dataset fingerprint before any number is shown.
3. **Read the summary**
   Expected total yield, number of consignments flagged for attention and the collection point groupings.
4. **Review one record**
   The officer opens a consignment and sees the yield estimate, the attention probability and the cluster it belongs to.
5. **Decide and act**
   The officer accepts the advice, overrides it with a reason, or asks for a second opinion. Every choice is logged.
6. **Learn next round**
   Accepted and overridden decisions are compared against the actual delivered weight at the end of the day.

Where the model outputs enter the flow

- Stage 3 and 4 consume artifacts/data_report.json, regression_metrics.json, classification_metrics.json and clustering_metrics.json produced by run_all.py.
- Stage 4 calls the same prediction logic exposed by predict.py, so the number the officer sees is the number the command line returns for the same record.
- Stage 5 writes the officer's decision next to the model advice; that pairing is the audit trail.
- Stage 6 compares advice against delivered weight, which is how the team detects drift.

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Annotated wireframe - overview and data quality

Low-fidelity layout. Numbered annotations below each screen explain the reasoning.

SCREEN A - Dispatch overview

HarvestLink | Round 2026-09-30 AM | fingerprint 3ff48e11... | models v1.0.0

EXPECTED YIELD	NEEDS ATTENTION	COLLECTION POINT GROUPS
24,180 kg	14 of 92 consignments	2 groups found
+/- 1,900 kg	15%	k = 2, silhouette 0.14

Consignment table: record_id | plot_area_ha | distance_km | arrival_hour | expected yield | attention probability | group
R0012 | 1.80 | 31.4 | 15 | 21,450 kg | 0.91 HIGH | group 0

Filters: attention only | arrival after 12:00 | distance > 25 km
Sort: attention probability (descending)

Advisory only - the officer confirms every dispatch decision. Source: AI_A1_G15.csv (unmodified).

Annotations

1. The dataset fingerprint is always visible so any user can prove which file produced these numbers.
2. Percentages next to counts, because '14 of 92' is easier to judge than '14'.
3. Risk is sorted to the top of the table, so the officer starts where attention is needed.

SCREEN B - Data quality gate (shown before any prediction)

FILE ACCEPTED 260 rows | 9 columns | 6 features | column order matches schema
Fingerprint 3ff48e11e25a904ccc5f2ed7810a0959c1de1f280b095f562f032302183e9d94

WARNINGS 6 missing cells (5 rainfall_mm, 1 soil_ph) filled with training medians
1 exact duplicate row removed. 0 unexpected columns.

[Re-run pipeline] [Download data_report.json] [Continue with warnings]

Annotations

4. Green means the schema matched exactly; amber means the results are usable but caveated.
5. The user is told exactly what was changed and how many rows were affected, before any model runs.

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Annotated wireframe - prediction, warning and clusters

The record detail view combines all three model outputs for one consignment.

SCREEN C - Consignment detail

EXPECTED YIELD
17,318 kg
range 15,400 - 19,200 kg (approx.)

DISPATCH ATTENTION: ROUTINE
probability 0.47 (threshold 0.50)

Why: + distance_km (14.0 km) and + arrival_hour (09:00) push the score up; - rainfall_mm (81 mm) pulls it down.
Largest standardised contributions from the logistic regression model.

Group 0 of 2 - similarity distance 1.12 - typical profile: 1.4 ha, 96 mm rain, 15 km
A description of similar inputs, not a verified farming category.

[ACCEPT advice] [OVERRIDE - reason required] [ASK SUPERVISOR]
Override reason (required): [buyer price] [truck capacity] [local knowledge] [other]
Every choice is stored with the fingerprint, model version and timestamp.

Human decision required. The system never auto-dispatches.

Annotations

6. Uncertainty is stated as a range and a probability, never as a bare point estimate.
7. Feature contributions make the score explainable without exposing mathematics.
8. Cluster wording explicitly refuses to claim a real-world category.
9. Override is a first-class action with a mandatory reason, so disagreement becomes data.

SCREEN D - Collection point groups

Scatter of collection points
coloured by group
2 groups, silhouette 0.14
(weak separation - use with caution)

Group 0 n=171 | 1.4 ha | 15.2 km | 09:20
Group 1 n= 89 | 2.1 ha | 18.6 km | 11:40

Routes are proposed to the supervisor,
never assigned automatically.

Annotations

10. The weak silhouette score is surfaced in the interface instead of being hidden from the user.

Responsible AI states

A dashboard that looks confident when the model is not is worse than no dashboard. The four states below are designed as first-class screens, not as error messages.

State 1 - Uncertainty

The yield estimate is always presented as a range derived from the test-set residual spread, not as one exact number. The attention flag shows its probability and the threshold in use. When the probability sits between 0.35 and 0.50 the card turns amber and says 'close to the line - review advised' instead of quietly reporting 'routine'.

State 2 - Missing or unexpected data

The quality gate runs before any prediction. Missing cells are imputed with training medians and the affected columns are named. Unexpected columns are listed and ignored, never silently consumed. If a required column is absent, or the target column is missing, the dashboard refuses to predict and explains which column is needed.

State 3 - Model limitation

A permanent limitations panel states, in plain language, that the models were trained on one issued fictional dataset, that the regression is linear, that the clustering silhouette was weak, and that clusters are not verified categories. The footer of every screen shows the dataset fingerprint and model version so a decision can always be traced to a specific model.

State 4 - Human override

The officer can accept, override with a mandatory reason, or escalate. Overriding is never penalised. The interface never auto-dispatches. A weekly review compares advice against the weight actually delivered, which is the only honest way to find out whether the model is still useful.

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Visual system and rationale

Palette

Primary blue	#0072B2	yield estimate, normal state	contrast on white 5.19:1
Amber	#E69F00	review zone, warnings, missing data	contrast on white 2.25:1
Green	#009E73	confirmed group, quality gate passed	contrast on white 3.42:1
Vermillion	#D55E00	attention required, override	contrast on white 3.87:1
Purple	#CC79A7	model limitations, provenance	contrast on white 3.06:1

The palette is the Okabe-Ito colour-blind-safe set. Ratios above are computed, not estimated. Colour is never the only signal: every state also carries a text label, because roughly one in twelve men cannot reliably separate the amber and green used here.

Labels and legends

- Every number carries its unit: 'kg' for yield, 'km' for distance, 'probability' for the flag.
- Cluster labels are always written as 'group 0 of 2', which makes it obvious that the numbering is arbitrary and not a ranking.
- Bars and dots always carry a value label, so the chart still works if it is printed in greyscale.
- The dataset fingerprint and model version appear in the footer of every screen.

Three justified design decisions

Show the probability, not just the flag

A binary 'attention / routine' hides how close the decision was. In this scenario a false negative is the expensive error (spoiled produce, rejected delivery), so the officer needs to see that 0.47 is nearly 0.50 and can be re-checked cheaply. Showing the number converts a hidden risk into a visible one.

Publish the fingerprint and the quality gate on every screen

The assessor, the manager and the officer must all be able to prove which file produced a number. Putting the SHA-256 and the cleaning warnings in the interface makes provenance a routine part of the workflow rather than a forensic exercise after something goes wrong.

Make override a first-class button with a mandatory reason

Front-line staff hold local knowledge the model cannot see: a buyer's price change, a broken bridge, a truck that is already full. If overriding is hidden or punished, staff will work around the tool and the cooperative loses both the decision and the feedback signal. A required reason turns every disagreement into labelled data for the next training round.